

STATEMENT OF WORK

INSPECTION AND TESTING OF THE FIRE SPRINKLER SYSTEMS

GREAT PLAINS AREA INDIAN HEALTH SERVICE LOCATIONS:

BELCOURT, ND

EAGLE BUTTE, SD

FT. THOMPSON, SD

FT. YATES, ND

KYLE, SD

LOWER BRULE, SD

MOBRIDGE, SD

PINE RIDGE, SD

RAPID CITY, SD

ROSEBUD, SD

SISSETON, SD

WAGNER, SD

WANBLEE, SD

FEBRUARY 2025

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STATEMENT OF WORK

Inspection and Testing of the Fire Sprinkler Systems

Great Plains Area

**Belcourt & Ft. Yates, ND, Eagle Butte, Ft. Thompson, Kyle, Lower Brule,
Mobridge, Pine Ridge, Rapid City, Rosebud, Sisseton, Wagner & Wanblee, SD**

1. Project Description

1.1 Title: Inspection and Testing of the Fire Sprinkler Systems

1.2 Location: This service is to be provided at multiple locations: Belcourt & Ft. Yates, ND, Eagle Butte, Ft. Thompson, Kyle, Lower Brule, Mobridge, Pine Ridge, Rapid City, Rosebud, Sisseton, Wagner & Wanblee, SD.

1.3 Period of Performance: Quarterly fire sprinkler inspections must be conducted within 30 days of the Notice to Proceed.

1.4 Description: The purpose of this contract is to provide quarterly inspections and tests of the fire sprinkler systems at various facilities in the Great Plains Area Indian Health Service, which includes locations in North Dakota and South Dakota. It is essential to perform these inspections and tests as required by code to maintain continued accreditation from the Centers for Medicare and Medicaid Services and the Joint Commission for our facilities.

1.5 Background: The purpose of fire sprinkler systems is primarily to provide a reasonable degree of protection for life and property from fire through standardization of design, installation, and testing requirements for sprinkler systems, including private fire service mains, based on sound engineering principles, test data, and field experience.

1.6 Related Documents

- 1.6.1** This SOW and any other standard government or IHS documents are included and considered a part of this contract.
- 1.6.2** NFPA 13, Standard for the Installation of Sprinkler Systems, 2010 Edition
- 1.6.3** NFPA 25, Standard for the Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems, 2011 Edition
- 1.6.4** NFPA 72, National Fire Alarm Code, 2010 Edition
- 1.6.5** NFPA 99, Standard for Healthcare Facilities, 2012 Edition
- 1.6.6** NFPA 101, Life Safety Code, 2012 Edition

1.7 Common Acronyms or Definitions

- 1.7.1** IHS – Indian Health Service
- 1.7.2** CMS – Centers for Medicare & Medicaid Services
- 1.7.3** CO – Contracting Officer
- 1.7.4** COR – Contracting Officer’s Representative
- 1.7.5** PO – Project Officer
- 1.7.6** IGCE – Independent Government Cost Estimate
- 1.7.7** RFI – Request For Information

2. Scope of Work (SOW): Perform quarterly inspections of the fire sprinkler systems located at selected Great Plains Area facilities.

The contractor shall provide all material, labor, equipment, and supervision required to accomplish the following:

2.1 General.

- 2.1.1** Inspection, testing, and maintenance shall be performed by qualified personnel.
- 2.1.2** The contractor shall notify the fire department and the alarm-receiving facility before testing or shutting down a system or its supply.
 - 2.1.2.1** The notification of system shutdown or test shall include the purpose for the shutdown or test, the system or component involved, the estimated time of shutdown or test, and the expected duration of the shutdown or test.
 - 2.1.2.2** The fire department and the alarm-receiving facility shall be notified when the system, supply, or component is returned to service or when the test is complete.
- 2.1.3** **Water Supply Status.** During inspection, testing, and maintenance, water supplies, including fire pumps, shall remain in service unless under constant attendance by qualified personnel.
- 2.1.4** **Inspection.** System components shall be inspected at intervals specified in NFPA 25, 2011 edition.
- 2.1.5** **Testing.**
 - 2.1.5.1** All components and systems shall be tested to verify that they function as intended.
 - 2.1.5.2** The frequency of tests shall be in accordance with NFPA 25, 2011 edition.
 - 2.1.5.3** Fire protection system components shall be restored to full operational condition following testing, including reinstallation of plugs and caps for auxiliary drains and test valves.
 - 2.1.5.4** Test results shall be compared with those of the original acceptance test (if available) and with the most recent test results.
- 2.1.6** **Maintenance.** Maintenance shall be performed to keep the system equipment operable or to make repairs.
- 2.1.7** **Safety.**
 - 2.1.7.1** **General.** Inspection, testing, and maintenance activities shall be conducted in accordance with applicable safety regulations.
 - 2.1.7.2** **Confined Spaces.** Legally required precautions shall be taken before entering confined spaces such as tanks, valve pits, or trenches.
 - 2.1.7.3** **Fall Protection.** Legally required equipment shall be worn or used to prevent injury from falls to personnel.
 - 2.1.7.4** **Hazardous Materials.** Legally required equipment shall be used where working in an environment with hazardous materials present.
 - 2.1.7.5** **Electrical Safety.** Legally required precautions shall be taken when testing or maintaining electric controllers for motor-driven fire pumps.

2.2 Sprinkler Systems.

- 2.2.1** **Inspection.**
 - 2.2.1.1** **Sprinklers.**
 - 2.2.1.1.1** Sprinklers shall be inspected from the floor level annually.
 - 2.2.1.1.1.1** Sprinklers shall not show signs of leakage; shall be free of corrosion, foreign materials, paint, and physical damage; and shall be installed in the correct orientation.
 - 2.2.1.1.1.2** Sprinklers installed in concealed spaces such as above suspended ceilings shall not require inspection.

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2.2.1.2 Pipe and Fittings. Sprinkler pipes and fittings shall be inspected annually from floor level.

- 2.2.1.2.1** Pipe and fittings shall be in good condition and free of mechanical damage, leakage, and corrosion.
- 2.2.1.2.2** Sprinkler piping shall not be subjected to external loads by materials either resting on the pipe or hanging from the pipe.
- 2.2.1.2.3** Pipe and fittings installed in concealed spaces such as above suspended ceilings shall not require inspection.

2.2.1.3 Hangers and Seismic Braces. Sprinkler pipe hangers and seismic braces shall be inspected annually from the floor level.

- 2.2.1.3.1** Hangers and seismic braces shall not be damaged, loose, or unattached.
- 2.2.1.3.2** Hangers and seismic braces installed in concealed spaces such as above suspended ceilings shall not require inspection.

2.2.1.4 Gauges.

- 2.2.1.4.1** Gauges on wet pipe and deluge sprinkler systems shall be inspected quarterly to ensure that they are in good condition and that normal water supply pressure is being maintained.
- 2.2.1.4.2** Gauges on dry and preaction systems shall be inspected quarterly to ensure that normal air or nitrogen and water pressures are being maintained.
- 2.2.1.4.3** For dry pipe or preaction systems protecting freezers with two air pressure gauges on the airline (s) between the compressor and the dry pipe or preaction valve, the air pressure gauge near the compressor shall be compared quarterly to the pressure gauge above the dry pipe or preaction valve.

2.2.1.5 Waterflow Alarms and Supervisory Signal Initiating Devices. Waterflow alarm and supervisory signal initiating devices shall be inspected quarterly to verify that they are free of physical damage.

2.2.1.6 Hydraulic Design Information Sign. The hydraulic design information sign shall be inspected quarterly to verify that it is provided, attached securely to the sprinkler riser, and is legible.

2.2.1.7 Information Sign. The information sign shall be inspected annually to verify that it is provided, securely attached, and legible.

2.2.1.7.1 Each sign shall be secured with a corrosion-resistant wire, chain, or other approved means.

2.2.1.7.1.1 Each sign shall indicate the location of the area served by the system.

2.2.1.7.1.2 Each sign shall indicate the location of auxiliary drains and low-point drains for dry pipe and preaction systems.

2.2.1.7.1.3 Each sign shall indicate the presence and location of antifreeze or other auxiliary systems.

2.2.2 Testing.

2.2.2.1 Sprinklers.

2.2.2.1.1 Verify sprinklers in service for 50 years have been replaced or representative samples from one or more sample areas have been tested and then retested at 10-year intervals.

2.2.2.1.2 Verify sprinklers manufactured before 1920 have been replaced.

2.2.2.1.3 Verify sprinklers manufactured using fast-response elements that have been in service for 20 years have been replaced or representative samples have been tested and then retested at 10-year intervals.

2.2.2.1.4 Verify sprinklers in service for 75 years have been replaced or representative samples from one or more sample areas have been tested and retested at 5-year intervals.

2.2.2.1.5 Verify dry sprinklers that have been in service for 10 years have been replaced or representative samples have been tested and then retested at 10-year intervals.

2.2.2.2 Gauges.

2.2.2.2.1 Gauges shall be replaced every 5 years or tested every 5 years by comparison with a calibrated gauge.

2.2.2.2.2 Gauges not accurate within 3 percent of the full scale shall be recalibrated.

2.2.2.3 Waterflow Alarm Devices.

2.2.2.3.1 Mechanical waterflow alarm devices, including, but not limited to, water motor gongs, shall be tested quarterly.

2.2.2.3.2 Vane-type and pressure switch-type waterflow alarm devices shall be tested semiannually.

2.2.2.3.3 Testing waterflow alarm devices on wet pipe systems shall be accomplished by opening the inspector's test connection.

2.2.2.3.3.1 Where freezing weather conditions or other circumstances prohibit the use of the inspector's test connection, the bypass connection shall be permitted to be used.

2.2.2.3.4 Fire pumps shall not be taken out of service during testing unless constantly attended by qualified personnel.

2.2.2.3.5 Testing waterflow alarm devices on dry pipe, preaction, or deluge systems shall be accomplished by using the bypass connection.

2.2.2.4 Antifreeze Systems. Annually, before the onset of freezing weather, the antifreeze solution shall be tested.

2.2.2.4.1 Using installation records, maintenance records, information from the owner, chemical tests, or other reliable sources of information, the type of antifreeze in the system shall be determined.

- 2.2.2.4.2** The specific gravity of each solution shall be checked using a hydrometer with a suitable scale or refractometer having a scale calibrated for the antifreeze solution.
- 2.2.2.4.3** The antifreeze solution shall be tested at its most remote portion and where it interfaces with the wet pipe system.
- 2.2.2.4.4** When antifreeze systems have a capacity larger than 150 gallons, a test at one additional point for every 100 gallons shall be made.

2.3 Fire Pumps.

2.3.1 General.

2.3.1.1 Perform all applicable tasks specified in Table 8.1.2 of NFPA 25, 2011 edition.

2.3.2 Inspection. Perform a quarterly inspection of the pump assembly.

- 2.3.2.1** Verify the pump house heat is adequate, not less than 40°F for pump rooms with diesel engine-driven pumps with engine heaters.
- 2.3.2.2** Verify the pump house head is adequate, not less than 70°F for pump rooms with diesel engine-driven pumps without engine heaters.
- 2.3.2.3** Verify the pump house ventilating louvers are free to operate.
- 2.3.2.4** Verify pump suction, discharge, and bypass valves are fully open.
- 2.3.2.5** Verify pump system piping is free of leaks.
- 2.3.2.6** Verify that the suction line pressure gauge reading is within acceptable range.
- 2.3.2.7** Verify that the system line pressure gauge reading is within acceptable range.
- 2.3.2.8** Verify that the suction reservoir has the required water level.
- 2.3.2.9** Verify wet pit suction screens are unobstructed and in place.
- 2.3.2.10** Verify that waterflow test valves are in the closed position.
- 2.3.2.11** Verify the controller pilot light (power on) is illuminated.
- 2.3.2.12** Verify transfer switch normal pilot light is illuminated.
- 2.3.2.13** Verify that the isolating switch is closed.
- 2.3.2.14** Verify that the reverse phase alarm pilot light is off or the normal phase rotation pilot light is on.
- 2.3.2.15** Verify that the oil level in the vertical motor sight glass is within acceptable range.
- 2.3.2.16** Verify that the power-to-pressure maintenance (jockey) pump is provided.
- 2.3.2.17** Verify that the diesel engine fuel tank is at least two-thirds full.
- 2.3.2.18** Verify that the diesel engine controller selector switch is in the auto position.
- 2.3.2.19** Verify batteries' voltage readings are within acceptable range.
- 2.3.2.20** Verify that the batteries' charging current readings are within acceptable range.
- 2.3.2.21** Verify that the batteries' pilot lights are on or battery failure pilot lights are off.
- 2.3.2.22** Verify all alarm pilot lights are off.
- 2.3.2.23** Verify that the diesel engine running time meter is running.
- 2.3.2.24** Verify diesel engine oil level in right angle gear drive is within acceptable range.
- 2.3.2.25** Verify that the diesel engine crankcase oil level is within acceptable range.
- 2.3.2.26** Verify diesel engine cooling water is within acceptable range.
- 2.3.2.27** Verify electrolyte level in batteries is within acceptable range.
- 2.3.2.28** Verify battery terminals are free from corrosion.
- 2.3.2.29** Verify water-jacket heater is operating.
- 2.3.2.30** Verify that the steam pressure gauge reading is within acceptable range.

2.3.3 Testing.

2.3.3.1 Frequency. A quarterly non-flow test shall be conducted for fire pumps without recirculating water back to the pump suction.

2.3.3.2 No-Flow Condition.

- 2.3.3.2.1** A test of fire pump assemblies shall be conducted without flowing water.
- 2.3.3.2.2** The test shall be conducted by starting the pump automatically.
- 2.3.3.2.3** The electric pump shall run for a minimum of 10 minutes.
- 2.3.3.2.4** The diesel pump shall run for a minimum of 30 minutes.
- 2.3.3.2.5** A valve installed to open as a safety feature shall be permitted to discharge water.
- 2.3.3.2.6** Qualified personnel shall be in attendance whenever the pump is in operation.
- 2.3.3.2.7** Pertinent visual observations or adjustments shall be conducted while the pump is idle.
 - 2.3.3.2.7.1** Record the system suction and discharge pressure gauge readings.
 - 2.3.3.2.7.2** For pumps that use electronic pressure sensors to control the fire pump operation, record the current pressure and the highest and the lowest pressure shown on the fire pump controller event log.
- 2.3.3.2.8** Pertinent visual observations or adjustments shall be conducted while the pump is running.
 - 2.3.3.2.8.1** Record the pump starting pressure from the pressure switch or pressure transducer.
 - 2.3.3.2.8.2** Record the system suction and discharge pressure gauge readings.
 - 2.3.3.2.8.3** Inspect the pump packing glands for slight discharge.
 - 2.3.3.2.8.4** Adjust gland nuts if necessary.
 - 2.3.3.2.8.5** Inspect for unusual noise or vibration.
 - 2.3.3.2.8.6** Inspect packing boxes, bearings, or pump casing for overheating.
 - 2.3.3.2.8.7** Record the pressure switch or pressure transducer reading and compare it to the pump discharge gauge.
 - 2.3.3.2.8.8** For pumps that use electronic pressure sensors to control the fire pump operation, record the current pressure and the highest and the lowest pressure shown on the fire pump controller event log.
 - 2.3.3.2.8.9** For electric motor and radiator-cooled diesel pumps, check the circulation relief valve for operation to discharge water.
 - 2.3.3.2.8.10** Record the time for the motor to accelerate to full speed.
 - 2.3.3.2.8.11** Record the time controller is on the first step (for reduced voltage or reduced current starting).
 - 2.3.3.2.8.12** Record the time the pump runs after starting (for automatic stop controllers).
 - 2.3.3.2.8.13** Record the time for a diesel engine to crank.
 - 2.3.3.2.8.14** Record the time for a diesel engine to reach running speed.
 - 2.3.3.2.8.15** Record the diesel engine oil pressure gauge, speed indicator, water, and oil temperature indicators periodically while the engine is running.
 - 2.3.3.2.8.16** Record any abnormalities.

- 2.3.3.2.8.17** Inspect the diesel engine heat exchanger for cooling waterflow.
- 2.3.3.2.8.18** Record the steam pressure gauge reading.
- 2.3.3.2.8.19** Record the time for the steam turbine to reach running speed.

2.3.3.3 Annual Flow Testing.

2.3.3.3.1 An annual test of each pump assembly shall be conducted by qualified personnel under no-flow (churn), rated flow, and 150 percent of the pump-rated capacity flow of the fire pump by controlling the quantity of water discharged through approved test devices.

2.3.3.3.1.1 If available suction supplies do not allow flowing of 150 percent of the rated pump capacity, the fire pump shall be tested to the maximum allowable discharge.

2.3.3.3.1.2 The annual test shall be conducted by one of the following methods:

2.3.3.3.1.2.1 Use of Pump Discharge via Hose Streams.

2.3.3.3.1.2.1.1 Pump suction and discharge pressures and the flow measurements of each hose stream shall determine the total pump output.

2.3.3.3.1.2.1.2 Care shall be taken to prevent water damage by verifying there is adequate drainage for the high-pressure water discharge from hoses.

2.3.3.3.1.2.2 Use of Pump Discharge via Bypass Flowmeter to Drain or Suction Reservoir. Pump suction and discharge pressures and the flowmeter measurements shall determine the total pump output.

2.3.3.3.1.2.3 Use of Pump Discharge via Bypass Flowmeter to Pump Suction (Closed-Loop Metering).

2.3.3.3.1.2.3.1 Pump suction and discharge pressures and the flowmeter measurements shall determine the total pump output.

2.3.3.3.1.2.3.2 When testing includes recirculating water back to the fire pump suction, the temperature of the recirculating water shall be monitored to verify that it remains below temperatures, which could result in equipment damage as defined by the pump and engine manufacturers.

- 2.3.3.3.1.3** Where the annual test is conducted periodically in accordance with 2.3.3.3.1.2.3 (Use of Pump Discharge via Bypass Flowmeter to Pump Suction), a test shall be conducted every 3 years in accordance with 2.3.3.3.1.2.1 (Use of Pump Discharge via Hose Streams) or 2.3.3.3.1.2.2 (Use of Pump Discharge via Bypass Flowmeter to Drain or Suction Reservoir) instead of the method described in 2.3.3.3.1.2.3 (Use of Pump Discharge via Bypass Flowmeter to Pump Suction).
- 2.3.3.3.1.4** Where 2.3.3.3.1.2.2 (Use of Pump Discharge via Bypass Flowmeter to Drain or Suction Reservoir) or 2.3.3.3.1.2.3 (Use of Pump Discharge via Bypass Flowmeter to Drain or Suction Reservoir) is used, the flowmeter shall be adjusted immediately before conducting the test in accordance with the manufacturer's instructions.
 - 2.3.3.3.1.4.1** If the test results are not consistent with the previous annual test, 2.3.3.3.1.2.1 (Use of Pump Discharge via Hose Streams) shall be used.
 - 2.3.3.3.1.4.2** If testing in accordance with 2.3.3.3.1.2.1 (use of Pump Discharge via Hose Streams) is not possible, a flowmeter calibration shall be performed, and the test shall be repeated.
- 2.3.3.3.2** Inspect the circulation relief valve for operation to discharge water at no-flow condition (churn).
- 2.3.3.3.3** Inspect the pressure relief valve (if installed) for proper operation.
- 2.3.3.3.4** Record the electric motor voltage and current (all lines) at each flow condition.
- 2.3.3.3.5** Record the pump speed in rpm at each flow condition.
- 2.3.3.3.6** Record the simultaneous (approximately) readings of pump suction and discharge pressures and pump discharge flow at each flow condition.
- 2.3.3.3.7** For electric motor-driven pumps, do not shut down the pump until it has run for 10 minutes.
- 2.3.3.3.8** For diesel motor-driven pumps, do not shut down the pump until it has run for 30 minutes.
- 2.3.3.3.9** For installations having a pressure relief valve, the operation of the relief valve shall be closely observed during each flow condition to determine where the pump discharge pressure exceeds the normal operating pressure of the system components.
 - 2.3.3.3.9.1** The pressure relief valve shall also be observed during each flow condition to determine whether the pressure relief valve closes at the proper pressure.
 - 2.3.3.3.9.2** The pressure relief valve shall be closed during flow conditions if necessary to achieve minimum rated characteristics for the pump and reset to normal position after the pump test.
 - 2.3.3.3.9.2.1** When it is necessary to close the relief valve to achieve minimum rated characteristics for the pump, the pump discharge control valve shall be closed if the pump churn pressure exceeds the system-rated pressure.

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- 2.3.3.6.5** Voltage reading at the motor within 5 percent below or 10 percent above the rated (i.e., nameplate) voltage shall be considered acceptable.
- 2.3.3.6.6** The pump performance shall be evaluated using the unadjusted flow rates and pressures to ensure the pump can supply the system demand as supplied by the owner.

2.4 Valves, Valve Components, and Trim.

2.4.1 General Provisions.

- 2.4.1.1** All pertinent personnel, departments, or agencies shall be notified that testing or maintenance of the valve and associated alarms is to be conducted.
- 2.4.1.2** Verify all system valves are protected from physical damage and accessible.
- 2.4.1.3** Before opening a test or drain valve, it shall be verified that adequate provisions have been made for drainage.
- 2.4.1.4** Main Drain Test. A main drain test shall be conducted annually for each water supply lead-in to a building water-based fire protection system to determine whether there has been a change in the condition of the water supply.
 - 2.4.1.4.1** In systems where the sole water supply is through a backflow preventer and/or pressure-reducing valves, the main drain test of at least one system downstream of the device shall be conducted every quarter.
 - 2.4.1.4.2** When there is a 10 percent reduction in full flow pressure when compared to the original acceptance test or previously performed tests, the cause of the reduction shall be identified and corrected if necessary.

2.4.2 Control Valves in Water-Based Fire Protection Systems.

- 2.4.2.1** Verify each control valve is identified and has a sign indicating the system or portion of the system it controls.
- 2.4.2.2** Verify systems that have more than one control valve that must be closed to work on a system have a sign on each affected valve referring to the existence and location of other valves.
- 2.4.2.3** Verify each normally open valve is secured using a seal lock or is electrically supervised in accordance with the applicable NFPA standards.
- 2.4.2.4** Verify each normally closed valve is secured using a seal lock or is electrically supervised in accordance with the applicable NFPA standards.
- 2.4.2.5 Inspection.**
 - 2.4.2.5.1** All valves shall be inspected quarterly.
 - 2.4.2.5.1.1** Verify all valves are in the normal open or closed position.
 - 2.4.2.5.1.2** Verify all valves are sealed, locked, or supervised.
 - 2.4.2.5.1.3** Verify all valves are accessible.
 - 2.4.2.5.1.4** Verify post indicator valves (PIVs) are provided with correct wrenches.
 - 2.4.2.5.1.5** Verify all valves are free from external leaks.
 - 2.4.2.5.1.6** Verify all valves are provided with applicable identification.
- 2.4.2.6 Testing.**
 - 2.4.2.6.1** Each control valve shall be operated annually through its full range and returned to its normal position.
 - 2.4.2.6.2** Post indicator valves shall be opened until spring or torsion is felt in the rod, indicating that the rod has not become detached from the valve.
 - 2.4.2.6.2.1** This test shall be conducted every time the valve is closed.
 - 2.4.2.6.3** Post indicator and outside screw and yoke valves shall be backed a one-quarter turn from the fully open position to prevent jamming.

- 2.4.2.6.4 A main drain test shall be conducted any time the control valve is closed and reopened at the system riser.
- 2.4.2.6.5 **Supervisory Switches.**
 - 2.4.2.6.5.1 Valve supervisory switches shall be tested semiannually.
 - 2.4.2.6.5.1.1 A distinctive signal shall indicate movement from the valve's normal position during either the first two revolutions of a hand wheel or when the stem of the valve has moved one-fifth of the distance from its normal position.
 - 2.4.2.6.5.1.2 The signal shall not be restored at any valve position except the normal position.
- 2.4.2.7 **Maintenance.**
 - 2.4.2.7.1 Operate and lubricate the stems of the outside screw and yoke valves annually.
 - 2.4.2.7.2 The valve then shall be completely closed and re-opened to test its operation and distribute the lubricant.
- 2.4.3 **System Valves.**
 - 2.4.3.1 **Alarm Valves.**
 - 2.4.3.1.1 **Inspection.** Alarm valves and system riser check valves shall be inspected quarterly.
 - 2.4.3.1.1.1 Verify gauges indicate normal supply water pressure is being maintained.
 - 2.4.3.1.1.2 Verify valves are free of physical damage.
 - 2.4.3.1.1.3 Verify valves are in the appropriate open or closed position.
 - 2.4.3.1.1.4 Verify the retarding chamber or alarm drains are not leaking.
 - 2.4.3.1.2 Alarm valves and their associated strainers, filters, and restriction orifices shall be inspected internally every 5 years.
 - 2.4.3.2 **Check Valves.**
 - 2.4.3.2.1 **Inspection.** Valves shall be inspected internally every 5 years to verify that all components operate correctly, move freely, and are in good condition.
 - 2.4.3.3 **Preaction Valves and Deluge Valves.**
 - 2.4.3.3.1 **Inspection.**
 - 2.4.3.3.1.1 Verify valve enclosures for preaction and deluge valves subject to freezing are maintained at a minimum temperature of 40°F.
 - 2.4.3.3.1.2 Low-temperature alarms, if installed in valve enclosures, shall be inspected annually at the beginning of the heating season to verify that they are free of physical damage.
 - 2.4.3.3.1.3 The preaction or deluge valve shall be externally inspected quarterly.
 - 2.4.3.3.1.3.1 Verify valves are free from physical damage.
 - 2.4.3.3.1.3.2 Verify all trim valves are in the appropriate open or closed position.
 - 2.4.3.3.1.3.3 Verify the seat valve is not leaking.
 - 2.4.3.3.1.3.4 Verify electrical components are in service.
 - 2.4.3.3.1.4 The interior of the preaction or deluge valve and the condition of detection devices shall be inspected annually when the trip test is conducted.

- 2.4.3.3.1.4.1 Internal inspection of valves that can be reset without removal of a faceplate shall be conducted every 5 years.
 - 2.4.3.3.1.5 Strainers, filters, restricted orifices, and diaphragm chambers shall be inspected internally every 5 years.
 - 2.4.3.3.2 **Testing.**
 - 2.4.3.3.2.1 The priming water level in supervised preaction systems shall be tested quarterly for compliance with the manufacturer's instructions.
 - 2.4.3.3.2.2 Except for preaction systems protecting freezers, preaction valves shall be trip tested every 3 years with the control valve fully open.
 - 2.4.3.3.2.3 During those years, when the full flow testing with the control valve fully open is not required, the preaction valve shall be trip tested with the control valve partially open.
 - 2.4.3.3.2.4 Preaction or deluge valves protecting freezers shall be trip tested in a manner that does not introduce moisture into the piping in the freezer.
 - 2.4.3.3.2.5 Preaction systems shall be tested once every 3 years for air leakage using one of the following test methods:
 - 2.4.3.3.2.5.1 A pressure test at 40 psi for 2 hours. The system shall be permitted to lose up to 3 psi during the duration of the test.
 - 2.4.3.3.2.5.2 With the system at normal system pressure, shut off the air source (compressor or shop air) for 4 hours. If the low air pressure alarm goes off within the period, the air leaks shall be addressed.
 - 2.4.3.3.2.6 Manual Operation. Manual actuation devices shall be operated annually.
 - 2.4.3.3.2.7 Return to Service. After the full flow test, the system shall be returned to service by the manufacturer's instructions.
 - 2.4.3.3.2.8 Grease or other sealing materials shall not be applied to the seating surfaces or preaction or deluge valves.
 - 2.4.3.3.2.9 Low air pressure alarms, if provided, shall be tested quarterly by manufacturer's instructions.
 - 2.4.3.3.2.10 Low-temperature alarms, if installed in valve enclosures, shall be tested annually at the beginning of the heating season.
 - 2.4.3.3.2.11 Automatic air pressure maintenance devices, if provided, shall be tested yearly at the time of the annual preaction or deluge valve trip test in accordance with the manufacturer's instructions.
 - 2.4.3.3.3 **Maintenance.**
 - 2.4.3.3.3.1 During the annual trip test, the interior of the preaction or deluge valve shall be cleaned thoroughly, and the parts replaced or repaired as necessary.

2.4.3.3.3.1.1 Interior cleaning and parts replacement or repair shall be every 5 years for valves that can be reset without removal of a faceplate.

2.4.3.3.3.2 Auxiliary drains in preaction or deluge systems shall be operated after each system operation.

2.4.3.4 Dry Pipe Valves/Quick-Opening Devices.

2.4.3.4.1 Inspection.

2.4.3.4.1.1 Valve enclosures for dry pipe valves subject to freezing shall be inspected quarterly during cold weather to verify a minimum temperature of 40°F.

2.4.3.4.1.2 Low-temperature alarms, if installed in valve enclosures, shall be inspected annually at the beginning of the heating season to verify that they are free of physical damage.

2.4.3.4.1.3 Verify systems with auxiliary drains shall have a sign at the dry or preaction valve indicating the number of auxiliary drains and the location of each drain.

2.4.3.4.1.4 The dry pipe valve shall be externally inspected quarterly.

2.4.3.4.1.4.1 Verify the valve is free of physical damage.

2.4.3.4.1.4.2 Verify all trim valves are in the appropriate open or closed position.

2.4.3.4.1.4.3 Verify the intermediate chamber is not leaking.

2.4.3.4.1.5 The interior of the dry pipe valve shall be inspected annually when the trip test is conducted.

2.4.3.4.1.6 Strainers, filters, and restricted orifices shall be inspected internally every 5 years.

2.4.3.4.2 Testing.

2.4.3.4.2.1 The priming water level shall be tested quarterly.

2.4.3.4.2.2 Each dry pipe valve shall be trip tested annually during warm weather.

2.4.3.4.2.2.1 Dry pipe valves protecting freezers shall be trip-tested in a manner that does not introduce moisture into the piping in the freezers.

2.4.3.4.2.2.2 Every 3 years, the dry pipe valve shall be trip tested with the control valve fully open and the quick-opening device, if provided, in service.

2.4.3.4.2.2.3 During those years when full flow testing with the control valve fully open is not required, each dry pipe valve shall be trip tested with the control valve partially open.

2.4.3.4.2.2.4 Verify the air supply is capable of restoring normal air pressure in the system within 30 minutes.

2.4.3.4.2.2.5 Verify the air supply is capable of restoring normal air pressure in the system within 60 minutes in refrigerated spaces maintained below 5°F.

2.4.3.4.2.3 Grease or other sealing materials shall not be applied to the seating surfaces of dry pipe valves.

- 2.4.3.4.2.4** Quick-opening devices, if provided, shall be tested quarterly.
- 2.4.3.4.2.5** A tag or card that shows the date on which the dry pipe valve was last tripped and the name of the person and organization conducting the test shall be attached to the valve.
- 2.4.3.4.2.6** Low air pressure alarms, if provided, shall be tested quarterly in accordance with the manufacturer's instructions.
- 2.4.3.4.2.7** Low-temperature alarms, if installed in valve enclosures, shall be tested annually at the beginning of the heating season.
- 2.4.3.4.2.8** Automatic air pressure maintenance devices, if provided, shall be tested annually during the dry pipe valve trip test in accordance with the manufacturer's instructions.
- 2.4.3.4.2.9** Dry pipe systems shall be tested once every 3 years for gas leakage using one of the following test methods:
 - 2.4.3.4.2.9.1** A pressure test at 40 psi for 2 hours. The system shall be permitted to lose up to 3 psi during the duration of the test.
 - 2.4.3.4.2.9.2** With the system at normal system pressure, shut off the air source (compressor or shop air) for 4 hours. If the low air pressure alarm goes off within the period, the air leaks shall be addressed.

2.4.3.4.3 Maintenance.

- 2.4.3.4.3.1** During the annual trip test, the interior of the dry pipe valve shall be cleaned thoroughly, and parts replaced or repaired as necessary.
- 2.4.3.4.3.2** Auxiliary drains in dry pipe sprinkler systems shall be drained after each operation of the system.

2.4.4 Pressure-Reducing Valves and Relief Valves.

2.4.4.1 Inspection and Testing of Sprinkler Pressure-Reducing Valves.

- 2.4.4.1.1** All valves shall be inspected quarterly.
 - 2.4.4.1.1.1** Verify valves are in the open position.
 - 2.4.4.1.1.2** Verify valves are not leaking.
 - 2.4.4.1.1.3** Verify valves are maintaining downstream pressures in accordance with the design criteria.
 - 2.4.4.1.1.4** Verify valves are in good condition, with handwheels installed and unbroken.
- 2.4.4.1.2** A full flow test shall be conducted on each valve at 5-year intervals and shall be compared to the previous test results.
 - 2.4.4.1.2.1** Adjustments shall be made in accordance with the manufacturer's instructions.
- 2.4.4.1.3** A partial flow test adequate to move the valve from its seat shall be conducted annually.

2.4.5 Backflow Prevention Assemblies.

2.4.5.1 Inspection.

- 2.4.5.1.1** The isolation valves on double-check assemblies (DCA) and double-check detector assemblies (DCDA) shall be inspected quarterly to ensure that the valves are in the normal open position.

2.4.5.1.2 The isolation valves on reduced-pressure assemblies (RPA) and reduced-pressure detector assemblies (RPDA) shall be inspected quarterly to ensure that the valves are in the normal open position.

2.4.5.1.2.1 RPAs and RPDAs shall be inspected quarterly to ensure that the differential-sensing valve relief port is not continuously discharging.

2.4.5.1.3 Backflow prevention assemblies shall be inspected internally every 5 years to verify that all components operate correctly, move freely, and are in good condition.

2.4.5.2 Testing.

2.4.5.2.1 All backflow preventers installed in fire protection system piping shall be exercised annually by conducting a forward flow test at a minimum flow rate of the system demand.

2.4.5.2.1.1 The forward flow test shall not be required where annual fire pump testing causes the system flow rate to flow through the backflow preventer device.

2.4.5.2.2 Where hydrants or inside hose stations are located downstream of the backflow preventer, the forward flow test shall include hose stream demand.

2.4.5.2.3 Where connections do not permit verification of the forward flow test at the minimum flow rate of system demand, tests shall be conducted at the maximum flow rate possible.

2.4.6 Fire Department Connections.

2.4.6.1 Fire department connections shall be inspected quarterly.

2.4.6.1.1 Verify the fire department connections are visible and accessible.

2.4.6.1.2 Verify couplings or swivels are not damaged and rotate smoothly.

2.4.6.1.3 Verify plugs or caps are in place and undamaged.

2.4.6.1.4 Verify gaskets are in place and good condition.

2.4.6.1.5 Verify identification signs are in place.

2.4.6.1.6 Verify the check valve is not leaking.

2.4.6.1.7 Verify the automatic drain valve is in place and operating properly.

2.4.6.1.8 Verify the fire department connection clapper(s) is in place and operating properly.

2.4.6.1.9 Verify the interior of the connection is free of obstructions.

2.4.6.1.9.1 Any obstructions that are present shall be removed.

2.5 Internal Piping Condition and Obstruction Investigation.

2.5.1 Assessment of Internal Condition of Piping. An assessment of the internal condition of piping shall be conducted at a minimum of every 5 years.

2.5.2 Ice Obstruction. Dry pipe or preaction sprinkler system piping that protects or passes through freezers or cold storage rooms shall be inspected internally on an annual basis for ice obstructions at the point where the piping enters the refrigerated area.

3. Period of Performance.

3.1 The quarterly inspection tests shall be performed within 30 days from the Notice to Proceed.

3.2 Following the testing, a copy of each report shall be provided to the Project Officer for documentation purposes. This documentation shall be provided no later than 20 workdays after the completion of the final site visit and shall be in an organized, legible format. See Section 6 for additional information.

4. Contractor Responsibilities

- 4.1** Contractor personnel performing the testing/inspections shall possess appropriate credentials from a certifying organization, such as the National Fire Protection Association (NFPA 101, 99, 13, 25, 72), NFPA ITM and WBITM certification (NFPA 25), and National Institute for Certification in Engineering Technologies (NICET) Level II or III certification in "Inspection and Testing of Water-Based Systems". Supporting documentation for credentials shall be submitted with the bid proposal.
- 4.2** Following notice to proceed, the contractor shall submit a proposed inspection schedule to the Project Officer for approval. The schedule shall include facility locations, the date of planned inspections, and the name of each technician to perform each inspection. The contractor shall notify the Project Officer immediately concerning any scheduling changes. The contractor shall notify each Facility Manager a minimum of 72 hours before performing a service.
- 4.3** Failed component replacement and repairs will not be included within the contractor's responsibilities unless the failure of components is the result of the contractor's activities. However, the contractor is invited to provide recommendations resulting from the contracted service to the Facility Manager and the Project Officer. Any repair, service, or part not included within this contract will require a separate procurement from the local Service Unit before the repair, service, or part can be performed, installed, or ordered. Invoices without a corresponding procurement number will not be paid.

5. Government Responsibilities

- 5.1** The Facility Manager shall provide access to all necessary areas of the facility.
- 5.2** Pre-Construction Meeting: One Pre-Construction meeting shall be conducted before the start of the construction of the contract.

6. Records/Data

- 6.1** Following completion of the testing, reports shall be completed for each facility. All testing data shall be recorded on the attached applicable "RECORD OF INSPECTION, TESTING, & MAINTENANCE" forms or another form approved by the Project Officer. Reports shall be electronically completed utilizing a fillable PDF form provided by IHS. A copy of each report shall be provided to the Project Officer for documentation purposes. This documentation shall be provided no later than 20 workdays following the completion of the final site visit.
- 6.2** The contractor shall submit one package, after all sites have been visited, containing one invoice that includes all the locations to the Project Officer itemizing the service performed at each site. The invoice shall include the contract number, facility name, type of service performed at each facility, date of service for each facility, and payment amount per facility. The reports shall accompany the invoice. Individual invoices shall not be submitted to the locations within the contract.

7. Quality Assurance

- 7.1** Facility Managers will submit vendor evaluations to the Project Officer upon completion of each site visit, see attached sample. Successful contract completion will be determined by the timely submission of schedules, reports, and invoices. Accuracy of schedule and information in reports will also be taken into consideration. Thorough completion of services as indicated in the contractor tasks, vendor evaluations, and in general will also be considered when evaluating the successful completion of the contract.

8. Deliveries or Performance

- 8.1** Place of Performance/Hours of Operation

- 8.1.1** Sites to receive service are Belcourt (ND), Eagle Butte (SD), Ft. Thompson (SD), Ft. Yates (ND), Kyle (SD), Lower Brule (SD), Mobridge (SD), Pine Ridge (SD), Rapid City (SD), Rosebud (SD), Sisseton (SD), Wagner (SD), and Wanblee (SD).

8.1.1.1 Belcourt, ND

8.1.1.1.1 Hospital

8.1.1.1.2 Mental Health Clinic

8.1.1.2 Eagle Butte, SD

8.1.1.2.1 Hospital

8.1.1.3 Ft. Thompson, SD

8.1.1.3.1 Health Center

8.1.1.3.2 Substance Abuse Center

8.1.1.3.3 Wellness Center

8.1.1.4 Ft. Yates, ND

8.1.1.4.1 Hospital

8.1.1.4.2 Dental Clinic

8.1.1.4.3 Optometry/Mental Health Clinic

8.1.1.5 Kyle, SD

8.1.1.5.1 Health Center

8.1.1.5.2 Four Directions Clinic

8.1.1.5.3 Wellness Center

8.1.1.6 Lower Brule, SD

8.1.1.6.1 Health Center

8.1.1.6.2 Dental Clinic

8.1.1.7 Mobridge, SD

8.1.1.7.1 Treatment Center

8.1.1.8 Pine Ridge, SD

8.1.1.8.1 Hospital

8.1.1.8.2 Diabetes Education Building

8.1.1.8.3 LaCreek Clinic (Martin, SD)

8.1.1.9 Rapid City, SD

8.1.1.9.1 Health Center

8.1.1.10 Rosebud, SD

8.1.1.10.1 Hospital

8.1.1.11 Sisseton, SD

8.1.1.11.1 Health Center

8.1.1.12 Wagner, SD

8.1.1.12.1 Health Center

8.1.1.13 Wanblee, SD

8.1.1.13.1 Health Center

- 8.1.2** Inspection and testing shall be completed between the hours of 8:00 AM and 5:00 PM, M-F, unless otherwise approved by the Project Officer. No testing and maintenance shall be performed on a national holiday unless approved by the Project Officer.

8.2 Delivery Dates/Period of Performance

8.2.1 Quarterly inspection tests shall be performed within 30 days from the Notice to Proceed.

9. Payments to the Contractor

9.1 The Government will pay the Contractor a fixed price, as indicated below, which will constitute full compensation for all Contractor professional and incidental services, travel expenses, and materials necessary to furnish the complete project as required and specified in this SOW.

9.2 Progress payments will be paid following acceptance by the Government of that particular schedule item and a properly submitted and approved invoice.

10. IHS Personnel

Contracting Officer's Representative (COR)

Stottin Newbrough will be designated as the Contracting Officer's Representative (COR) for this Order. His address and telephone are:

Stottin Newbrough
Great Plains Area Indian Health Service
Division of Facilities Management & Engineering
3200 Canyon Lake Road
Rapid City, SD 57702
Office: 605-945-5452
Stottin.Newbrough@ihs.gov

Any change in COR designation will be documented by the Contracting Officer in writing.

Project Officer (PO)

Kevin Bliese is the Project Officer for this contract. His contact information is:

Kevin Bliese
Project Officer
Great Plains OEHE
Cell #: 605-310-7969
Kevin.Bliese@ihs.gov

11. Invoice Format

Invoice Processing Platform (IPP)

The Indian Health Service (IHS) is in the process of implementing an electronic invoicing system. *In compliance with the [Office of Management and Budget \(OMB\) M-15-19](#) memorandum, Improving Government Efficiency and Saving Taxpayer Dollars Through Electronic Invoicing, directing Federal agencies to adopt electronic invoicing as the primary means to disburse payment to vendors. Invoices submitted under any award resulting from this solicitation will be required to utilize the Invoice Processing Platform (IPP) in accordance with HHSAR 352.232-71, Electronic Submission and Processing of Payment Requests.*

IPP is a secure, web-based electronic invoicing system provided by the U.S. Department of the Treasury's Bureau of the Fiscal Service in partnership with the Federal Reserve Bank of St. Louis (FRSTL). Respondents to this solicitation are encouraged to register an account with IPP if they have not already done so. If your

organization is already registered to use IPP, you will not be required to re-register- however, we encourage you to make sure your organization and designated IPP user accounts are valid and up to date.

The IPP website address is: <https://www.ipp.gov>.

If you require assistance registering or IPP account access, please contact the IPP Helpdesk at (866) 973-3131 (M-F 8 AM to 6 PM ET) or IPPCustomerSupport@fiscal.treasury.gov.

Should the contractor feel that use of the IPP would be unduly burdensome, their response to this solicitation should include an explanation of this position for a determination by the contracting officer.

12. Authority

- 12.1** No person other than the Contracting Officer has the authority to bind the Government with respect to this contract.
- 12.2** No action or omission of any government employee or representative other than the Contracting Officer shall increase or decrease the scope of this contract or shall otherwise modify the terms and conditions of this contract.
- 12.3** In no event shall any of the following be effective or binding on the Government or imputed to the Contracting Officer with respect to this contract:
 - 12.3.1** An understanding or agreement between the Contractor and anyone other than the Contracting Officer;
 - 12.3.2** A purported modification or change order issued by anyone other than the Contracting Officer;
 - 12.3.3** A promise by anyone other than the Contracting Officer to provide additional funding or make payments or
 - 12.3.4** An order, direction, consent, or permission from anyone other than the Contracting Officer to:
 - 12.3.4.1** Incur costs in excess of a specified estimated cost, allotment of funds, or other ceiling or
 - 12.3.4.2** Expend hours in excess of a specified level of effort.

Appendix A – Fire Sprinkler Equipment Information and Location

Appendix B – Bid Schedule**INSPECTION AND TESTING OF THE FIRE SPRINKLER SYSTEMS
FOR THE GREAT PLAINS AREA INDIAN HEALTH SERVICE**

Location	Building	Cost
Belcourt	Hospital	
	Mental Health Clinic	
Eagle Butte	Hospital	
Ft. Thompson	Health Center	
	Substance Abuse	
	Wellness Center	
Ft. Yates	Hospital	
	Dental Clinic	
	Optometry Clinic	
Kyle	Health Center	
	Four Directions Clinic	
	Wellness Center	
Lower Brule	Health Center	
	Dental Clinic	
Mobridge	Treatment Center	
Pine Ridge	Hospital	
	Lacreek Clinic	
	Diabetes Education	
Rapid City	Health Center	
Rosebud	Hospital	
Sisseton	Health Center	
Wagner	Health Center	
Wanblee	Health Center	
	Total Bid:	

Total prices must include testing equipment, labor, overhead, profit, delivery, mobilization and demobilization, and all applicable taxes and fees (federal, state, local, and Tribal) for all fire sprinkler systems at IHS Service Unit locations listed in the work statement.

Appendix C – Inspection and Testing Schedule

Location	1 st Quarter Inspection/Test Date	2 nd Quarter Inspection/Test Date	3 rd Quarter Inspection/Test Date	4 th Quarter Inspection/Test Date
Belcourt				
Eagle Butte				
Ft. Thompson				
Ft. Yates				
Kyle				
Lower Brule				
Mobridge				
Pine Ridge				
Rapid City				
Rosebud				
Sisseton				
Wagner				
Wanblee				

*** Facility Manager to be notified 72 hours before performing a service.**

Appendix D – Report of Inspection, Testing, & Maintenance Form

Appendix E – Contract Performance Evaluation

Great Plains Area Fire Sprinkler Inspection Performance Evaluation

To be completed by the Facility Manager after inspection

Facility Name: _____

Date:_____ Contractor Rep onsite:_____

1.	Overall Evaluation	Poor	Average	Good	Excellent
2.	Performance	Poor	Average	Good	Excellent
3.	Quality of Services	Poor	Average	Good	Excellent
4.	Workmanship	Poor	Average	Good	Excellent
5.	Knowledge	Poor	Average	Good	Excellent

Comments:This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Signature of person completing evaluation: